

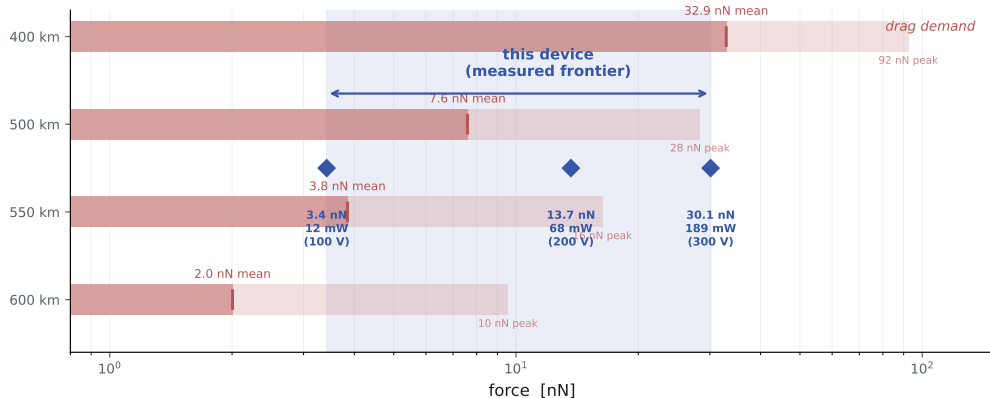
Propellantless Drag Compensation by Ambient-Electron Exchange

Measured PIC frontier pre-registered predictions zero propellant

R. S. C.

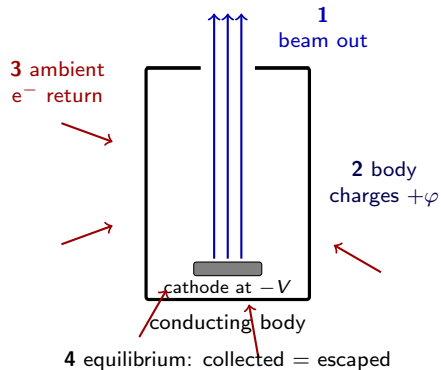
August 2026

Drag is nanonewtons — no flight thruster reaches that low



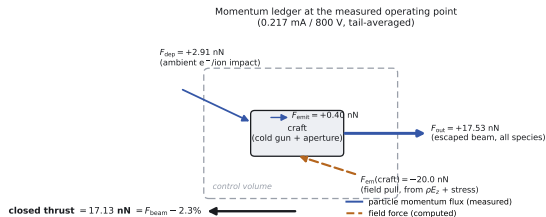
- ▶ LEO drag at 500–600 km: **2–8 nN mean** for a kg-class CubeSat.
- ▶ Smallest flight EP (FEEP/electrospray): $5\text{--}30\text{ }\mu\text{N}$, $\geq 8\text{ W}$ — $10^2\text{--}10^3\times$ above the demand.
- ▶ Dry-system floor (tank + feed + valves + PPU) $\approx 0.3\text{--}1\text{ kg}$: a large share of a kilogram-class CubeSat.

You already fly this device — as a neutralizer



1. Supply holds cathode at $-V$; electrons accelerate out through the aperture.
 2. Body charges positive ($+\varphi$).
 3. Ionosphere returns electrons to the skin until $I_{\text{collected}} = I_{\text{escaped}}$.
 4. **The ionosphere is both propellant tank and neutralizer.**
- **Hardware:** an HV supply, a cold gun, a hole.
 - Every ion/Hall thruster already carries the gun — run it alone and it *is* this thruster.

“The current comes back — doesn’t the momentum cancel?”



Full-return null test: body pinned above beam energy, escape = 0%.

Impact channel: -12.1 nN .

Closed thrust: $-0.39 \text{ nN} \approx 0$. ✓

At the operating points:

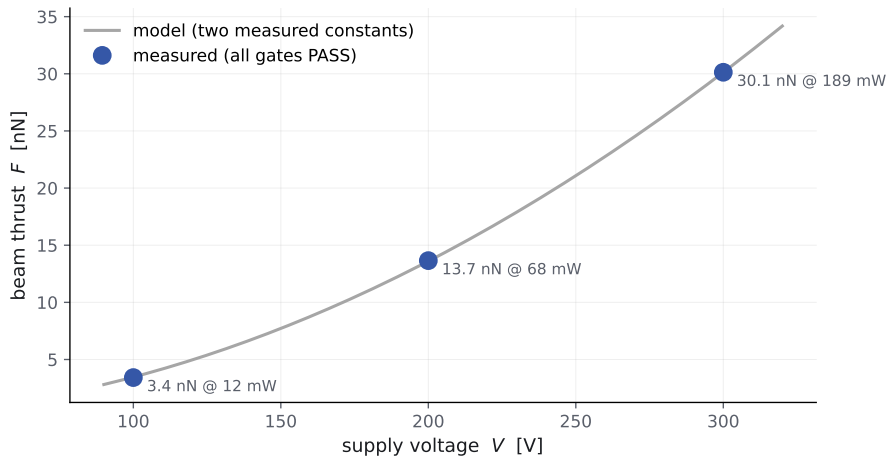
Return-channel momentum vs beam thrust:

1.0% at 300 V (0.30 vs 30.1 nN)

8.8% at 100 V

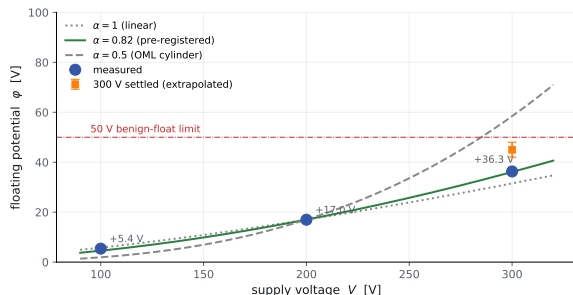
Physics ceiling: collected e^- arrive isotropically at $\sim e\phi$ vs directed exhaust at $\sim 210 \text{ eV}$ — even fully anti-directed arrival cancels $\leq 41\%$.

The frontier is measured, not projected



- ▶ Three production runs across the full 100–300 V hardware range; **all pre-registered acceptance gates PASS**.
- ▶ Thrust landed $<1\%$ from predictions recorded in the repository before the runs completed.

The collection law was decided by experiment



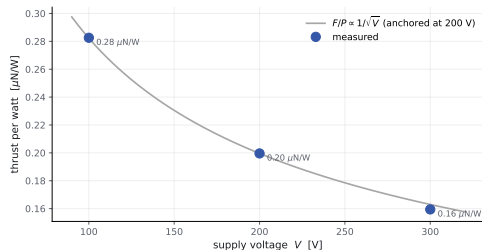
Collected current: $I = \beta A j_{\text{the}} (1 + \chi)^\alpha$,
 $\chi = e\phi/kT_e$.

Three exponents, all agreeing at 200 V, diverging at 300 V. Predictions recorded before the run:

law	predicted ϕ	verdict
$\alpha = 1$	31 V	refuted
$\alpha = 0.82$	36 V	survives
$\alpha = 0.5$	90 V	refuted
measured	36.3 V	

Fit across all three points: $\alpha = 0.85\text{--}0.89$.
Closes the analytical power model to 4–6%.

F/P is $200\times$ below your Hall thruster — by design



	this work	electrospray	Hall
η_{jet}	68–73 %	28–45 %	31–44 %
F/P [$\mu\text{N/W}$]	0.16–0.28	~ 5	~ 50

Both rows are true at once *because the exhaust is electrons*: superb energy-to-jet conversion at tiny momentum per watt.

The F/P penalty is **invisible at nN**, noticeable at μN , decisive at mN .

Below $\sim 0.1 \mu\text{N}$: us.

μN : electrospray.

mN: ion/Hall.

Complement, not competitor.

Density axis: the law is conservative

The $n_e/3$ continuation ($2.4\ \mu\text{s}$) is the campaign's only **fully settled float**:

- ▶ Pre-registered predictions: $\alpha=1$: 53 V; 0.82: 68 V; 0.5: 160 V.
- ▶ **Measured: +42.5 V, settled** ($-0.14\ \text{V}/\mu\text{s}$, flat to noise).
- ▶ $\alpha = 0.5$ refuted a second time.
- ▶ Every fixed exponent *overshoots* $\Rightarrow$ the fitted law is **conservative along density**.

Thin plasma does *not* end the envelope:

- ▶ 42.5 V clears the 50 V benign limit (a run this axis was expected to break).
- ▶ $F = 12.39\ \text{nN}$; escape 99.1 %; KE 122.0 vs 122.6 eV predicted.
- ▶ All acceptance gates PASS.

	anchor (n_0)	thin ($n_0/3$)
φ [V]	+17.0	+42.5
escape	98.4 %	99.1 %
settled?	no (800 ns)	yes ($2.4\ \mu\text{s}$)

Geomagnetic field: operating point untouched at flight strength

30 μT (flight strength):

- ▶ φ : 17.22 vs 16.98 V (no field).
- ▶ F : 13.64 vs 13.65 nN.
- ▶ Escape within 0.06 pp.
- ▶ Pre-registered null **holds on every bound**.
- ▶ Beam gyroradius $\geq 0.10\text{ m} \gg$ device.

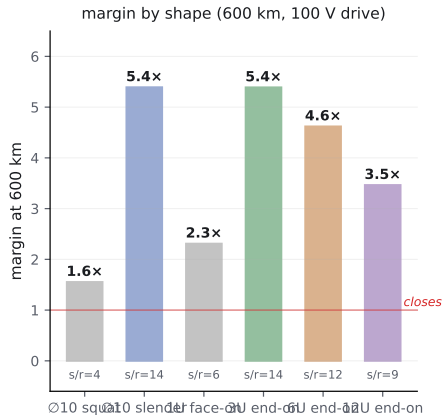
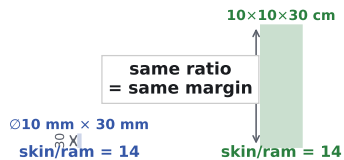
300 μT (10 $\times$, amplification):

- ▶ A collection tax through the float:
 $\varphi = +32.6\text{ V}$ (vs $+17.0\text{ V}$).
- ▶ Beam formation *unaffected*:
escape 98.3 %.
- ▶ The two-constant thrust law reproduces both:
13.56 / 12.03 nN predicted vs
13.64 / 12.06 nN measured.

The field acts on *collection* only — not on beam formation.

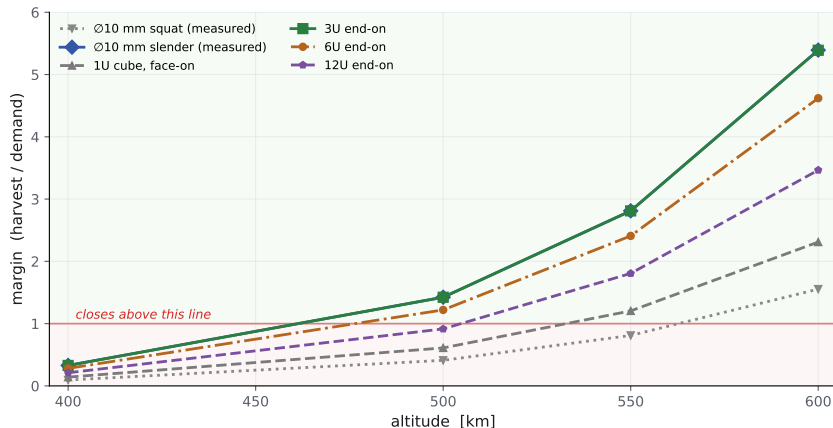
Size cancels: the corridor carries to CubeSats

100 × scale difference, identical feasibility



- ▶ Drag charges **ram silhouette**; collection and harvest pay from **skin**. Both areal $\Rightarrow$ **size cancels**.
- ▶ 3U at 600 km: 8.8 mA / 0.88 W. Bigger is *less* extrapolated.
- ▶ Caveat: CubeSat radii 25–60 λ_D (vs 2.5 measured) — different sheath regime, **not yet run**.

Mission corridor — with the failures stated plainly



- ▶ Year-long propagations, real F10.7/ A_p , per-5-min plasma + drag; minimum-power (V, I, φ) solved each step.
- ▶ **Closes at 550–600 km**; impulse-closes at 500 km; **does not close at 400 km** for any shape.
- ▶ Slender bodies: φ drops $17 \rightarrow 4.4$ V; thrust *rises* to 14.2 nN at same silhouette.

Where the model is still blind

Measured axes:

- ▶ **Voltage:** three-point frontier, all gates PASS, model closes to 4–6 %.
- ▶ **Density:** one $3\times$ step (settled, law conservative). Deeper night rows remain extrapolation.
- ▶ **Geometry:** slender body to $L/r = 6$; beyond that, open.
- ▶ **B field:** axial pair. Flight null holds; transverse far field cannot be posed in RZ — *open*.

Known approximations:

- ▶ Reduced ion mass ($400 m_e$ vs $O^+ 29\,166$).
- ▶ Stationary plasma (no ram wake).
- ▶ RZ symmetry (no 3D effects).
- ▶ Powers are beam power / V ; converter and cathode overheads not included.

Highest-value next run:

CubeSat sheath regime ($25\text{--}60 \lambda_D$).

Lower $\chi \Rightarrow$ faster settle.

Every row outside the measured envelope is flagged, never averaged in.

1. **The frontier is measured:** 3.4 nN @ 12 mW to 30 nN @ 189 mW, all gates PASS, predictions pre-registered.
2. **The thrust is real:** full-return null closes to zero; return-channel momentum $\leq 9\%$ of beam thrust.
3. **The collection law is discriminated on two axes** and is *conservative* along density; the geomagnetic field leaves the operating point untouched at flight strength.
4. **The mission corridor is honest:** closes at 550–600 km; fails at 400 km and says so.
5. **Size cancels:** a 3U CubeSat inherits the corridor unchanged.

Cheap to test on orbit: it's a gun and a hole.

Backup: spacecraft charging detail

- ▶ The float φ is the state variable: it rises until the ionosphere returns exactly the escaped current.
- ▶ The float robs the beam: accelerating potential is always $V - \varphi$ (charging tax: 5 % at 100 V, 12 % at 300 V).
- ▶ Gated benign ($\varphi \leq 50$ V) on every run; a choke detector aborts any run the ionosphere cannot neutralize.

V [V]	100	200	300
φ [V]	+5.4	+17.0	+36.3
escape	96.1 %	98.4 %	99.0 %
balance	1.5 %	3.2 %	3.5 %

Backup: the throttle principle

Ideal law (measured constants; near-unity escape, $\varphi \ll V$)

$$F = \sqrt{2m_e/e} I \sqrt{KE} \quad (\text{beam momentum flux})$$

$$P = VI \Rightarrow P = F\sqrt{V}/c_{\text{ideal}}, \quad c_{\text{ideal}} = \sqrt{2m_e/e}$$

throttle: largest feasible escaped current, lowest sufficient voltage

PIC: $c_F = 0.97 c_{\text{ideal}}$ (divergence), $\kappa = 0.81$ (space charge) $\Rightarrow 1.19\text{--}1.22\times$ the parameter-free bound

- ▶ Model: 11.7 / 65.8 / 177.9 mW vs measured 12.1 / 68.4 / 189 mW — within 4–6 %.
- ▶ Residual is the charging tax $V/(V - \varphi)$.

Backup: how we know (evidence stack)

Per run (all gated):

- ▶ Frozen, hashed physics config; manifest with seed, code version, git commit.
- ▶ 7–8 acceptance gates fixed *before* the run; zero waived.
- ▶ Ledger vs independent particle dumps: agreement at 10^{-9} (gate 2×10^{-2}).

Campaign-level:

- ▶ Validation ladder: thermal collection $\pm 1\%$ vs theory $\rightarrow$ biased $\rightarrow$ floating $\rightarrow$ gun rungs.
- ▶ Code-to-code parity: φ 0.05 %, escape 0.001 %, F 0.09 %.
- ▶ PPC $\times 2$ shifts $\leq 0.05\%$; grid (0.15 $\rightarrow$ 0.10 mm): F +4.0 %, KE +7.4 % — carried as the error band.

Backup: settle time and the 300 V late-slope caveat

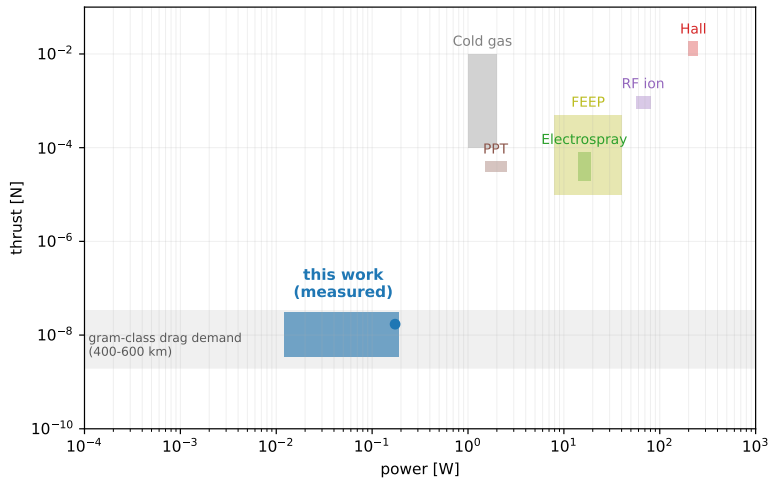
- ▶ Body $\approx$ capacitor charged by the escaping beam: $\tau \approx C\varphi/I_{\text{esc}} \approx 28$ ns; every run spans $\approx 29\tau$.
- ▶ At high χ the sheath equilibrates on the *ion* timescale: the 300 V late slope decayed $+44 \rightarrow +27$ mV/ns but was nonzero at 800 ns.
- ▶ Settled-value extrapolation: $\varphi_{\infty} \approx 42\text{--}48$ V — inside the 50 V gate.
- ▶ The $n_e/3$ continuation (2.4 μ s) is the only fully settled float.

- ▶ WarpX 26.5 (GPU), RZ electrostatic, embedded-boundary conductor with per-step charge-pump float update; particle reservoir (infinite-ionosphere recycle).
- ▶ Grid 0.15 mm ($\lambda_D = 1.97$ mm); CFL dt ; 16 ppc; 800 ns per production run; ~ 7 h on an RTX 3060.
- ▶ PPC $\times 2$ at the anchor: all 8 gates PASS. Grid 0.15 $\rightarrow$ 0.10 mm: $F +4.0\%$, KE $+7.4\%$.
- ▶ Reduced ion mass $400 m_e$: ladder-wide, disclosed.

Backup: “why not a tether?”

- ▶ Conceded: per watt, electrodynamic-tether thrust at km scale beats any reaction engine.
- ▶ For the bodies here the comparison inverts:
 - ▶ A CubeSat vs a deployed, stabilized conductor 10^3 – $10^6 \times$ its own length.
 - ▶ Thrust constrained to $/\mathbf{L} \times \mathbf{B}$ direction.
 - ▶ A bare tether still needs electron collection and emission at its ends — the same contactor physics this device is, minus the tether.
- ▶ Related physics is our foundation, not competitor: bare-tether OML collection (Sanmartín 1993), PMG plasma contactors (1993), E-sail electron gun (Janhunen).

Backup: F/P plane (incumbent context)



Incumbent ranges: NASA SoA Small Spacecraft Technology 2026; manufacturer datasheets (see paper Table 2).

Backup: propellant accounting

- ▶ “Propellant” = ambient electrons: $\dot{m} = (m_e/e) I \approx 2 \times 10^{-15} \text{ kg/s}$ at 0.342 mA $\approx$ **60 ng/year**.
- ▶ Effective $I_{sp} \sim 10^6 \text{ s}$ — not a merit figure (electrons are nearly free, momentum-poor mass); quoted only for EP bookkeeping.
- ▶ Five years of nN thrust $\approx 5 \text{ N s} \approx 0.3 \text{ g}$ electro spray propellant: the argument is never propellant *mass* — it is the **dry-system floor**.